AAE 51400 Intermediate Aerodynamics
3 credits

1 Course Information/Description

Thin airfoil theory and panel methods. Three-dimensional elementary solutions, lifting-line and lifting-surface models. Slender wing and slender body theory. Normal/oblique shock waves. One- and multi-dimensional flows and conical flows. Method of characteristics. Introduction to transonic flow, hypersonic flow, and airfoil design.

This course is designed to give the student exposure to a wide variety of aerodynamic problems and issues. It could be the first course in the area for a graduate student or an advanced course for an undergraduate. This course stresses fundamental ideas and concepts. It will be helpful for everybody interested in aerodynamics: the CFD user, the experimentalist, the practicing engineer.

2 Learning Resources and Reference Text

Brightspace Learning Management System:

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook:

There is no required textbook for this course.

3 Assignments

Homework will be assigned approximately every week and due one week later. The homework problems will test the students' understanding of the material presented during the lecture.

There will be two exams, a midterm exam and a final exam.

4 Course Learning Objectives

- Measure and differentiate between absolute, differential, and gauge pressure.
- Make basic pressure and velocity measurements.
- Utilize elementary flow visualization tools.
- Deduce fluid physics information from flow visualization.
- Calculate flow characteristics such as Reynolds number, friction factor, pressure and drag coefficient from laboratory measurements.

- Identify and discuss foundation-level fluid phenomena including boundary layers, wakes, laminar to turbulent transition, turbulence, flow separation, drag on a body, and surface pressure distributions.
- Design an experiment to meet a defined objective, conduct the experiment, analyze the data, and draw conclusions.

5 Evaluation

Homework assignments will account for 40% of the final grade, and the exams will account for 60% (40% for the exam with the higher score, 20% for the exam with the lower score). A percentage grade will be calculated, and final letter grades will be assigned according to the scale shown below (note that numerical scores will *not* be rounded). Typically, grades are not curved in this course.

Your course grade will be based on the following grading scale:

Letter Grade	Percentage
A+	98-100%
A	93-97.99%
A-	90-92.99%
B+	87-89.99%
B	83-86.99%
B-	80-82.99%
C+	77-79.99%
C	73-76.99%
C-	70-72.99%

Letter Grade	Percentage
D+	67-69.99%
D	63-66.99%
D-	60-62.99%
F	0-59.99%